

Quick Setup Guide

EdgeIPS™ 102 Rugged

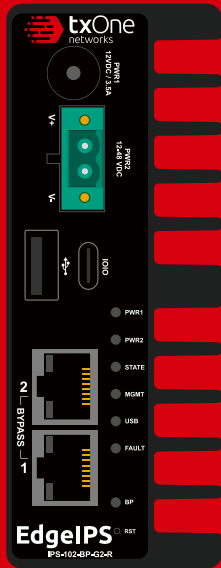


Support
hardware bypass
(Fail-Open only)

EdgeIPS 102 Rugged is a Transparent Security Box equipped with dual Ethernet LAN ports. Users can efficiently adapt the palm-sized EdgeIPS 102 Rugged into a flexible solution for the challenges of Ethernet communications.

1

Opening and Inspecting the Package



IPS-102-BP-G2-R-TX (Gen2)



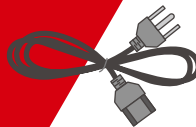
EdgeIPS 102 Rugged
Appliance



USB Dust Cover x1

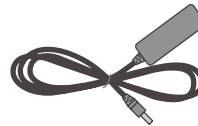


RJ45 Dust Covers x2



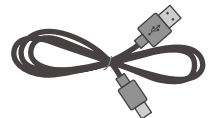
Power Cords x4
(US, UK, EU, JP)

*(The power cords are only
for this product)



Power Adapter x1

*(The power adapter is only
for this product)



Console Cable x1
(USB Type-C)



Terminal Block x1 &
DC in Jack to Terminal x1



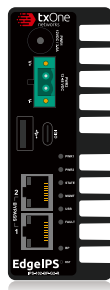
DIN Rail Kit x1
DIN Rail Screw M4x6L x2



Quick Setup Guide x1 &
Hardware Warranty Paper x1

2

Examining the EdgeIPS 102 Rugged



Front Panel

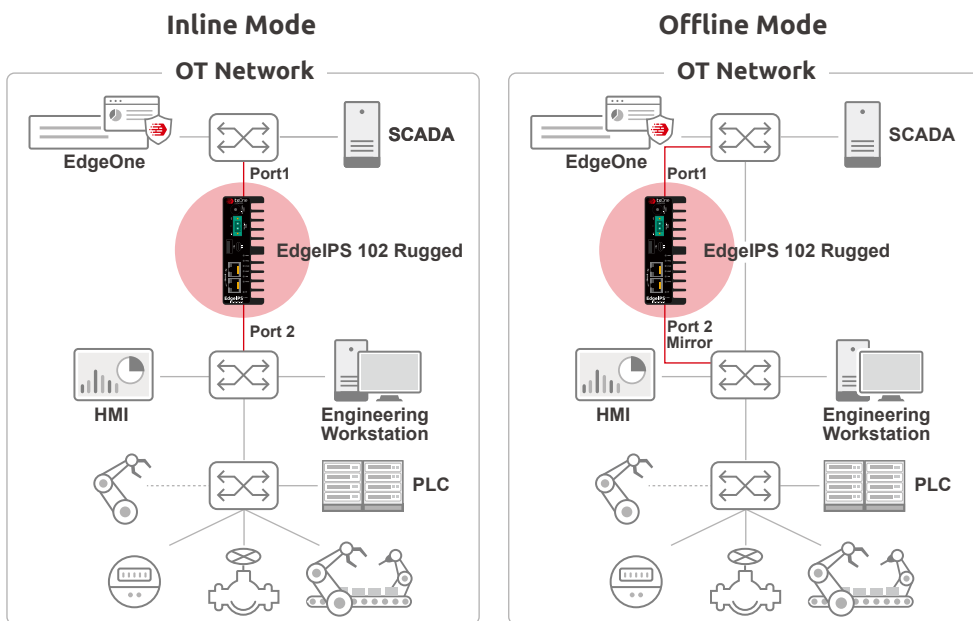


Back Panel

3 Deployment Checklist

Requirement	Details
Ethernet Cables	- Connect to port 1 to access web management console. - Connect switches or field assets to port 2.
Default IP Address of Web Console	Default web console access information: https://192.168.127.254  User name: admin Password: txone
Laptop with Ethernet Port (RJ-45 Type)	The computer must have one of the following browsers installed: - Microsoft Edge (version 15 or above) - Firefox (version 53 or above) - Safari (version 10.1 or above) - Google Chrome (version 63 or above)

4 Deployment Scenarios and System Operation Modes



Inline mode In inline mode, EdgeIPS 102 Rugged is deployed directly on the traffic path.

Offline mode In offline mode, EdgeIPS 102 Rugged is connected to a mirror port of network switch and mirrors traffic to itself.

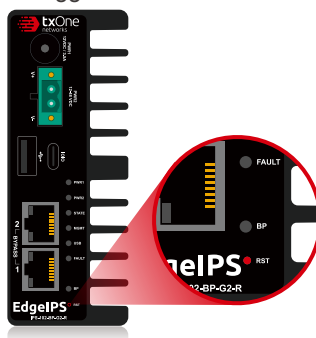
5 LED Indicators

The function of each LED is described in the table below.

PWR1	● Amber	Power 1 is on, and the device is functioning normally.
	● Off	Power 1 is off.
PWR2	● Amber	Power 2 is on, and the device is functioning normally.
	● Off	Power 2 is off.
STATE	● Green	EdgeOne sync is enabled and EdgeOne is connected.
	● Off	EdgeOne sync is disabled.
MGMT	● Green	EdgeOne sync is enabled and EdgeOne is connected.
	● Off	EdgeOne sync is disabled. Note: EdgelPS 102 Rugged will support EdgeOne sync if it is upgraded to V2.
USB	● Green	LED will be turned on when a USB flash drive is plugged in.
	● Alternating	A hardware or system fault occurs.
	● Blinking	USB data transmission is in progress.
	● Off	No USB flash drive is plugged in.
FAIL	● Amber	Hardware Bypass is enabled.
	● Off	Hardware Bypass is disabled.

6 Resetting to Factory Default Settings

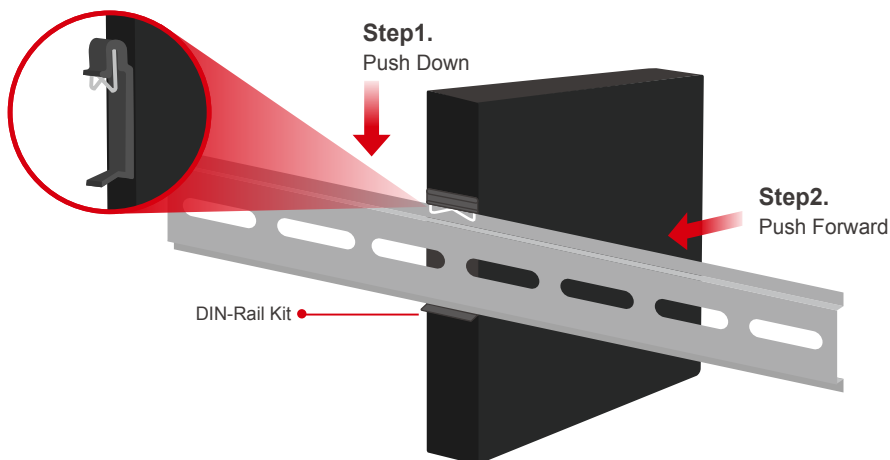
The reset button for the EdgelPS 102 Rugged is located on the front panel.



Reset Type	Description
Reboot System	Press and hold the reset button for 2 to 10 seconds. The MANAGED LED will begin to blink every one second, indicating the system is rebooting.
Restore Factory Default Settings	Press and hold the reset button for more than 10 seconds. The MANAGED LED will begin to blink every half second, indicating the system is resetting itself to factory defaults.

7 Mounting the EdgeIPS 102 Rugged to a DIN Rail

The aluminum DIN rail attachment plate should already be fixed to the back panel of EdgeIPS 102 Rugged when you take it out of the box. If you need to reattach the DIN rail attachment plate to EdgeIPS 102 Rugged, make sure the stiff metal spring is oriented towards the back, as shown in the following picture.

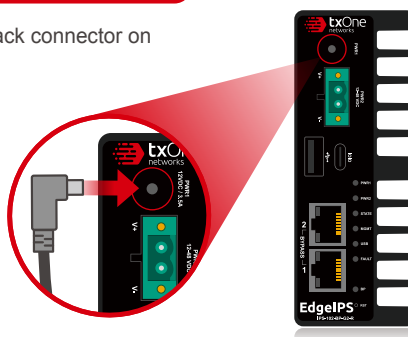


8 Wiring the Power Input

The EdgeIPS 102 Rugged has two types of power inputs. Use the provided power adapter (DC-IN Jack) or other 12V DC power supply (terminal block). The power input location and pin definition are shown in the following diagram.

Connecting the Power with a DC Power Adapter

Insert the provided adapter into the DC-IN jack connector on the front panel of EdgeIPS 102 Rugged.



Connecting the Power with Terminal Block

Step 1.

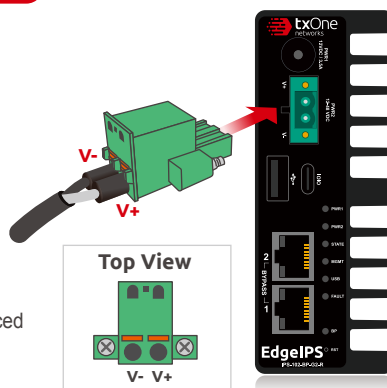
Open the terminal block's wire locks by pushing the buttons down.

Step 2.

Insert the positive and negative points of a DC wire into the V+ and V- terminals respectively.

Step 3.

When the wire is properly seated, the wire lock will be forced shut around it. Check whether the wire is properly fixed.

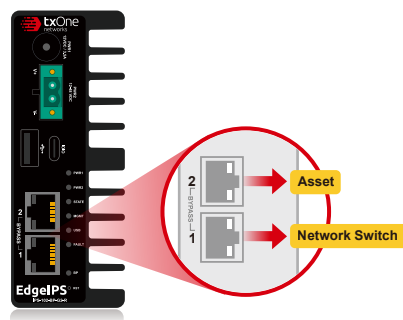
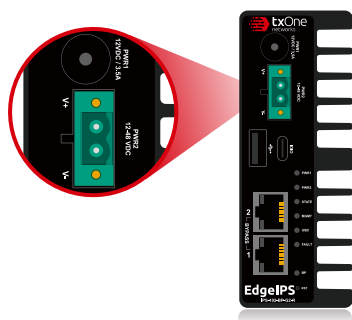


- ⚠ Consult with your electrician engineer before connecting EdgeIPS 102 Rugged to a DC power source.
- ⚠ Electrostatic discharge (ESD) can damage your EdgeIPS 102 Rugged.

9 Connecting the Power and Network

Connect a 12 to 48 VDC power line to the terminal block, or use the provided adapter to connect to the DC-IN jack; both methods supply power to the Edge device. If power is supplied correctly, the PWR1/P-WR2 LED will glow steadily. The power input location and pin definitions are shown in the following diagram.

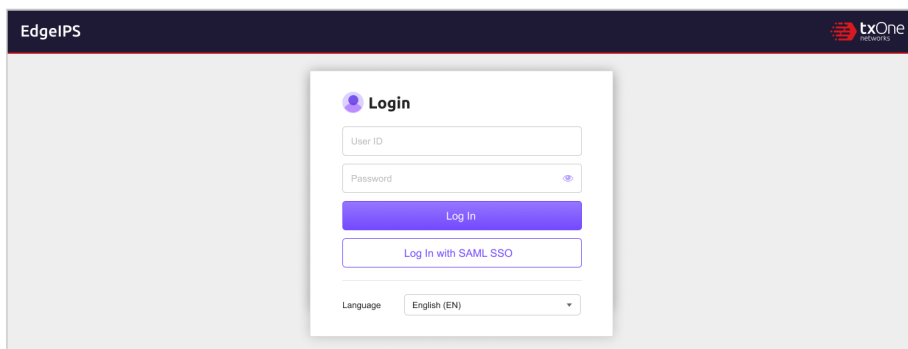
When deploying the Edge device to your OT network, make sure that the network switch is connected to Port 1 and the production assets are connected to Port 2.



10 Grounding the Unit

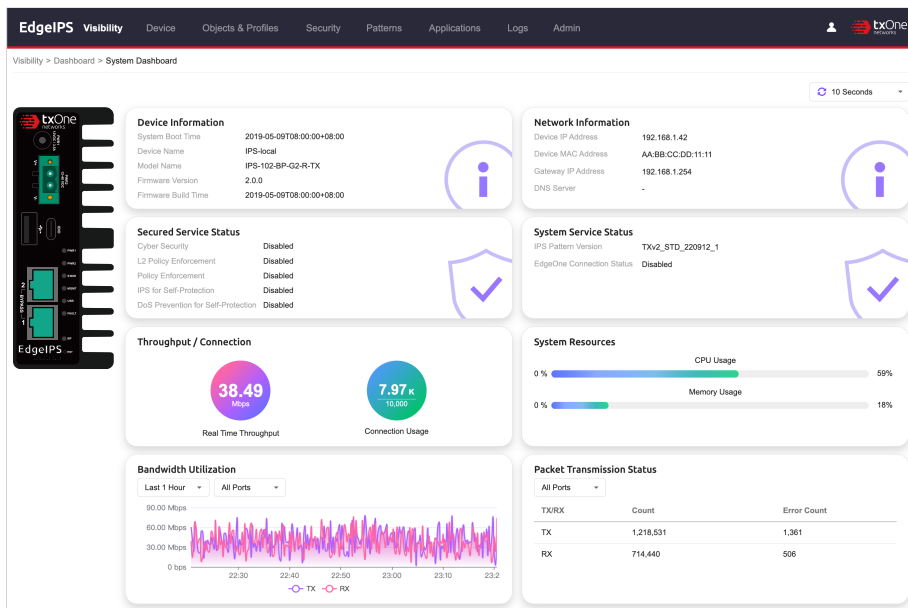
Grounding and wire routing help limit the effects of noise due to electromagnetic interference (EMI). Run the ground connection from the terminal block connector to the grounding surface prior to connecting the power. Please note that this product is intended to be mounted on a well-grounded mounting surface, such as a metal panel. The minimum cross-sectional area of the conductor must be equal to the input wiring cable's width/circumference.

11 Software Configuration



Before you connect the PC/laptop to EdgeIPS 102 Rugged, the PC's IP address should be set to 192.168.127.100. After that, connect the PC and EdgeIPS 102 Rugged using an Ethernet cable. Open a web browser on your PC and type "<https://192.168.127.254>" into the address bar. A pop-up window will be opened to ask for user name and password. Please type the default username, "admin", and password, "txone".

Now, the main screen will pop up so you can configure EdgeIPS 102 Rugged.



12 Configuring the Management Network Interface

1. Go to [Device] > [Network & Ports] > [Network Settings].

Network Settings

Connection Types

Single IP

Device IP Address*

192.168.1.42

Netmask*

255.255.255.0

Gateway*

192.168.1.1

DNS

8.8.8.8

☒ VLAN ID

42

LLDP Setting

Transmit LLDP

☒

2. In the [Network Settings] pane, configure the management network interface for EdgeIPS 102 Rugged.

Task	Action
Device IP Address	Configure Management IP Address of the EdgeIPS 102 Rugged.
Netmask	Configure Netmask of the EdgeIPS 102 Rugged.
Gateway	Configure Gateway Address of the EdgeIPS 102 Rugged.
DNS	Configure DNS setting of the EdgeIPS 102 Rugged.
Network VLAN ID	Configure Network VLAN ID of the EdgeIPS 102 Rugged.

13 Connecting to the Network

EdgeIPS 102 Rugged's ports 1 and 2 make up a network segment.

1. Port 1 is a "mixed" port design, functioning both as a management console port and a connecting port to a network node, so EdgeIPS 102 Rugged's port 1 can be connected either to your network switch/hub or to a laptop. Connecting this port to the network enables remote management of the device.
2. EdgeIPS 102 Rugged's port 2 links the IPS to an asset or small production zone you are targeting for protection.

 EdgeIPS 102 Rugged supports a fail-open hardware bypass mechanism, ensuring that network traffic continues to flow even if the device experiences a failure.

14 Checking the LED to Confirm Status

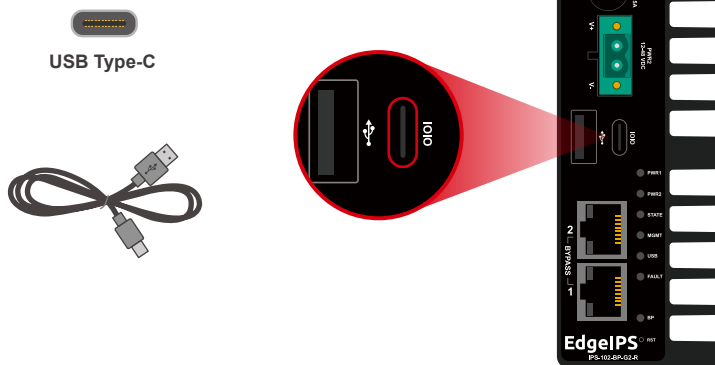
The LEDs on the Ethernet port show the deployment status.

	Off	The EdgeIPS 102 Rugged appliance cannot communicate with a network.
	Left: Green	GE LINK/ACT (off = no link, on = 1 Gbps link, blinking = activity)
RJ45	Right: Green	10/100Mbps LINK/ACT (off = no link, on = 10/100 Mbps link, blinking = activity)

Appendix: Connecting to the Console Port

The EdgeIPS 102 Rugged console port is a USB Type-C port located on the front panel of the case. It is designed for connecting to serial console terminals to view messages during a boot sequence or to debug system boot issues. To connect to a serial console terminal using a console cable, remove the protective cover from the port.

Serial Console Port, Cable, & Pinouts



The initial configuration for the EdgeIPS 102 Rugged using the command line interface (CLI) on a serial terminal client will use the following default settings:

- Baud Rate: 115200 bps
- Character Size: 8 bits
- Parity: None
- Stop Bits: 1
- Flow Control: None

Other Resources

Website

<https://www.txone.com/>

MyTXOne Portal

<https://my.txone.com/>

EdgeIPS Landing Page



EULA

<https://www.txone.com/legal/license-agreements/>

Contact Information

Email

 support@txone.com

Call Center

<https://help.txone.com/>

Phone

 US: +1 3465867975

 Netherland: +31 402310122

 Taiwan: +886 277275120



⚠ Warning

Ensure that the grounding screw is securely fastened and that the product is properly grounded.

- ⚠ Before removing and disposing of this product, ensure all data and configurations stored on the device are erased. You can do this by:
 1. manually resetting the product to factory settings, or
 2. handing the product to the TXOne support team, who will help you clear all stored data and configurations.
- ⚠ The power cord adapter should be connected to a socket outlet with an earthing connection.
- ⚠ The power cord and adapter must comply with Class II construction.
- ⚠ This product is intended to be supplied by a UL Listed Power Adapter with rated output 12-48Vdc, 7.5-1.87A (for Terminal Block) and 12Vdc, 7.5A (for DC Jack) and Tma 75°C min.
- ⚠ Do not replace batteries of the device. Risk of explosion may occur if the batteries are replaced with incorrect types.
- ⚠ Ne remplacez pas les piles de l'appareil. Un risque d'explosion peut survenir si les piles sont remplacées par des types incorrects.
- ⚠ Dispose of used batteries according to the instructions.
- ⚠ The equipment is intended to be installed in the restricted access location.
- ⚠ If using Class I adapter, power cord shall be connector to a socket-outlet with earthing connection.